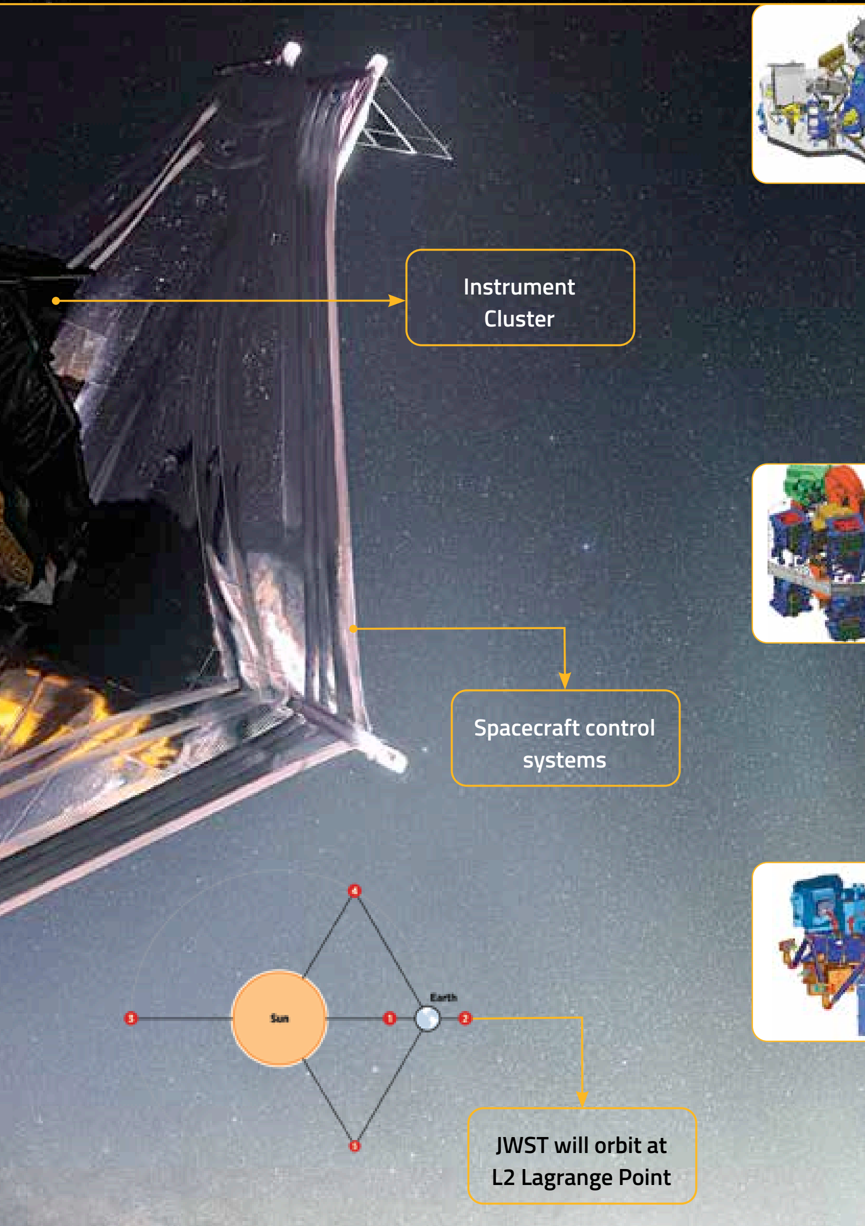
The James Webb Space Telescope aims to:

- Search for light from the earliest galaxies and stars that came right after the Big Bang
- Study how galaxies form and how they evolve
- Learn how planetary systems come to be and seek out the origins of life



Focal length	131.4 m
Number of primary mirror segments	18
Optical resolution	~0.1 arc-seconds
Wavelength coverage	0.6 - 28.5 microns
Size of sun shield	21.197 m x 14.162 m
Temp of sun shield layers	Layer 1 - 383K Max Layer 5 - 221K Max
Orbit	L2 Lagrange Point
Operating Temperature	under 50 K
Gold coating	1000 Angstroms spread over 25 m^2

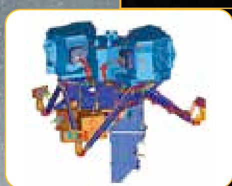
MAIN INSTRUMENTS

NEAR INFRARED SPECTROGRAPH (NIRSPEC)

Like the NIRCAM, the NIRSPEC will also cover the same wavelength range of 0.6 to 5 microns. It is a spectrograph which will be used to analyse the light captured from far-away objects by dispersing it over a spectrum. This way, the NIRSPEC can identify temperature, mass, chemical composition and more physical parameters of the object it's looking at. Each unique element present on that object can be identified this way.

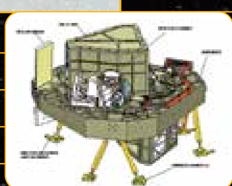
NEAR INFRARED CAMERA (NIRCAM)

The NIRCAM is the JWST's primary camera which is designed to cover the infrared wavelength between 0.6 to 5 microns. This will allow the NIRCAM to detect light from young stars in the Milky Way galaxy as well as from nearby galaxies. NIRCAM's coronagraphs will help see even the dim nearby objects which would have otherwise been drowned by bright objects in its vicinity.



MID-INFRARED INSTRUMENT (MIRI)

Operating over the wavelength range of 5 to 28 microns is the MIRI or Mid-Infrared Instrument. Redshited light from distant galaxies, newly formed stars, and faint objects from the Kuiper Belt can be detected by the sensitive equipment on the MIRI. The camera can capture breathtaking images thanks to its wide-field broadband imaging.



FINE GUIDANCE SENSOR / NEAR INFRARED IMAGER AND SLITLESS SPECTROGRAPH (FGS/NIRISS)

There's no point having the best cameras that money can buy if you can't point it in the right direction. That's where the FGS/NIRISS comes into the picture. It operates over a wave-length range between 0.8 to 5 microns and is a "guider" which helps point the telescope towards the right objective.